



TRINITY GRAMMAR SCHOOL
MATHEMATICS DEPARTMENT



YEAR 12 2009 ASSESSMENT TASK 2

MATHEMATICS

(2 UNIT/EXTENSION 1)

Time Allowed – *one hour*

(2 UNIT/EXTENSION 1)

Weighting 20% towards final result

Outcomes referred to: P2, P4, P5, P6, P7, P8, H1, H2, H4, H5, H6, H7, H8, H9

INSTRUCTIONS:

1. Attempt ALL questions.
 2. Show all necessary working.
 3. Begin each question on a new page.
 4. Each question is of equal value. Mark values are shown beside each part.
 5. Non-programmable silent Board of Studies approved calculators are permitted.
 6. If requested, additional writing sheets may be obtained from the examinations supervisor upon request.
 7. A double sided A4 page of notes is permitted to be referred to throughout this task.
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Question 1: A(2,-2), B(-2,-3) and C(0,2) are the vertices of a triangle ABC.

- (a)
- | | | |
|-------|---|---|
| (i) | Draw a sketch diagram of the triangle. | |
| (ii) | Find the length of AC and the gradient of AC. | 2 |
| (iii) | Find the equation of AC in the general form. | 2 |
| (iv) | Calculate the perpendicular distance of B from the side AC. | 2 |
| (v) | Hence, or otherwise, find the area of ΔABC . | 1 |
- (b) Find the co-ordinates of the point on the curve $y = 3x^2 - 2x + 1$ where the tangent is parallel to the straight line $4x - y - 1 = 0$.
- 3

Question 2:

- (a) For the curve $f(x) = x^4 - 4x^3$, find the values of x for which the curve is concave up.
- 2
- (b) Consider the curve $y = 3x^4 - 8x^3 + 6x^2$.
- | | | |
|-------|--|---|
| (i) | Find the co-ordinates of its stationary points. | 3 |
| (ii) | Use the second derivative to determine their nature. | 3 |
| (iii) | Find any points of inflexion. | 2 |
| (iv) | Sketch the curve for the domain $-1 \leq x \leq 2$ | 3 |

Question 3:

(a) Find the primitive functions of the following:

(i) $3x^5 - 2x^2 - x$

2

(ii) $3(3x-6)^4$

2

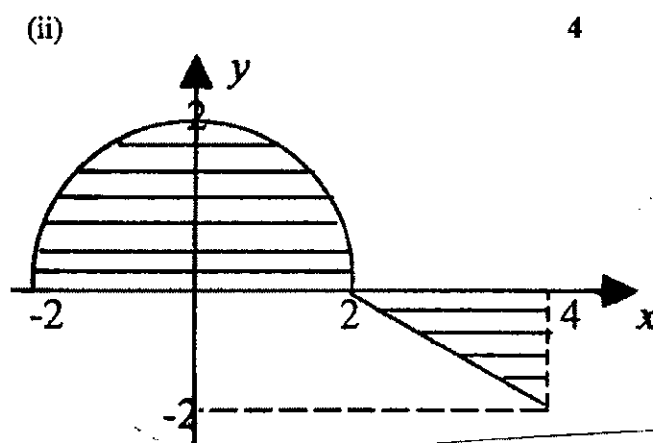
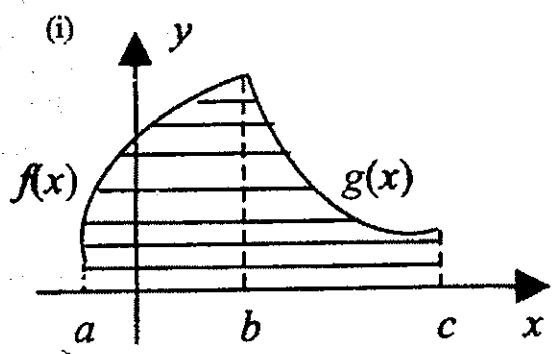
(iii) $\frac{2}{\sqrt{x}} + 2\sqrt{x} - 3$

3

(b) Express y in terms of x , given that $\frac{d^2y}{dx^2} = 12(x-1)^2$ and that at $x=1$, $\frac{dy}{dx} = 3$ and $y = 2$.

3

(c) Express the shaded area as the sum or difference of two integrals. DO NOT find the areas.



(d) Find the area enclosed between $y = x^3 - 6x^2 + 8x$ and the x -axis.

3

Question 4:

- (a) Find the volume of the solid generated when the semicircle $y = \sqrt{9 - x^2}$ is rotated about the x -axis.

3

- (b) (i) Show that the points A(0,3) and B(2,5) lie on the curve $y = 3 + \sqrt{2x}$.

2

- (ii) The part of the curve between A and B, is rotated about the y -axis. Find the volume of the solid generated.

4

Question 5:

- (a) Evaluate the definite integral $\int x(\sqrt{x} + x^2)dx$.

3

- (b) A rectangular pyramid is inscribed in a cylinder. The diameter of the base of the cylinder is 12cm and its height is 18cm. The sides of the rectangles are a cm and b cm.

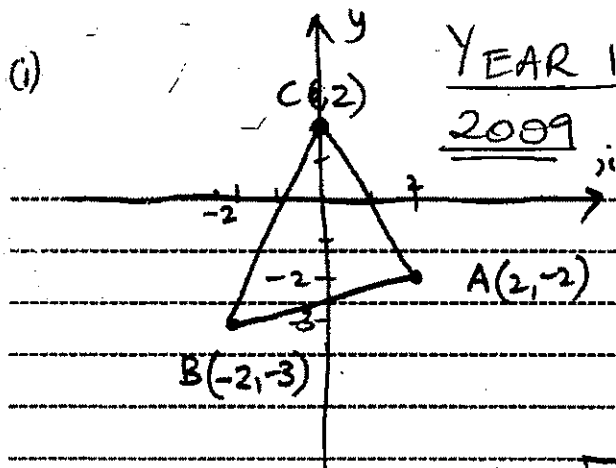
- (i) Show that the volume of the pyramid is $6\sqrt{144a^2 - a^4}$

2

- (ii) Show that the maximum volume of the pyramid and the volume of the cylinder are in the ratio $2:3\pi$.

4

Q1 (i)



YEAR 12: MATHEMATICS TASK 2

Student Number: SOLUTIONS2009

TO

(ii) Length AC $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
 $= \sqrt{(2 - 0)^2 + (-2 - 2)^2}$
 $= \sqrt{4 + 16}$

$$d = \sqrt{20} \text{ units}$$

✓ (1) $\sqrt{20} = 2\sqrt{5}$
 $\therefore d = 2\sqrt{5}$

Gradient AC $m = \frac{y_2 - y_1}{x_2 - x_1}$
 $= \frac{-2 - 2}{2 - 0}$
 $= \frac{-4}{2}$

$$\therefore m = -2$$

✓ (1)

(iii) Eqn AC $y - y_1 = m(x - x_1)$

$$y - -2 = -2(x - 2) \quad (1) \quad \checkmark$$

$$y + 2 = -2x + 4$$

$$2x + y - 2 = 0$$

(2) ✓

(iv) $D_p = \frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}}$

$$= \frac{|2 \cdot -2 + 1 \cdot -3 + -2|}{\sqrt{2^2 + 1^2}} \quad (1) \quad \checkmark$$

Point B(x, y) = (-2, -3)
 $A = 2$
 $B = 1$
 $C = -2$

$$= \frac{|-4 - 3 - 2|}{\sqrt{5}}$$

$$= \frac{|-9|}{\sqrt{5}}$$

$$= \frac{9}{\sqrt{5}} \text{ u}^2 \quad (2) \quad \checkmark$$

(v) $A = \frac{1}{2} \times b \times h$

$$= \frac{1}{2} \times \sqrt{20} \times \frac{9}{\sqrt{5}} \quad \checkmark$$

$$= \frac{1}{2} \times 2\sqrt{5} \times \frac{9}{\sqrt{5}}$$

$$A = 9 \text{ u}^2$$

(1)

$$Q1(b) \quad y = 3x^2 - 2x + 1$$

$$\text{Eqn of tangent } \frac{dy}{dx} = 6x - 2$$

$$\text{Straight line } y = 4x - 1$$

$$\text{here } m = 4 \quad (1) \checkmark$$

$$\therefore \text{ when } y' = 4,$$

that is the point where the gradients are parallel.

$$\therefore 4 = 6x - 2 \quad (1) \checkmark$$

$$x = 1 \quad \text{sub into } y = 3x^2 - 2x + 1.$$

$$y = 3 - 2 + 1$$

$$= 2$$

Point where tangent is parallel is $(1, 2)$ $(1) \checkmark$

(3)

QUESTION 2:

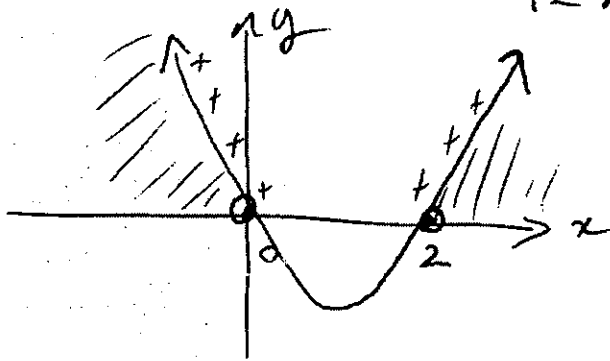
$$(a) \quad f'(x) = 4x^3 - 12x^2.$$

$$f''(x) = 12x^2 - 24x. \quad (1)$$

$\therefore$ For concave up, $f''(x) > 0$.

$$\therefore 12x^2 - 24x > 0$$

$$12x(x-2) > 0.$$



$$\therefore \begin{aligned} x &> 2 \\ x &< 0 \end{aligned}$$

(1)

$$(b) (i) \quad y = 3x^4 - 8x^3 + 6x^2.$$

$$\frac{dy}{dx} = 12x^3 - 24x^2 + 12x. \quad (1)$$

Let $\frac{dy}{dx} = 0$ for st. pts.

$$0 = 12x(x^2 - 2x + 1)$$

$$0 = 12x(x-1)(x-1)$$

$$\therefore x = 0, x = 1$$

when $x=0, y=0$ (1)
 $x=1, y=1$ (1)

(ii) $\frac{d^2y}{dx^2} = 36x^2 - 48x + 12$ (1)
 $\frac{d^2y}{dx^2} = 12(3x^2 - 4x + 1)$

Test $x=0$, $\frac{d^2y}{dx^2} = 12$ (1) $\cup$ min.

Test $x=1$, $\frac{d^2y}{dx^2} = 0$ (1) horizontal
 pt of inflex.

(iii) let $\frac{d^2y}{dx^2} = 0$

$$0 = 12(3x - 1)(x - 1)$$

$\therefore x=1$ and $x=\frac{1}{3}$. (1)

For $x=1$

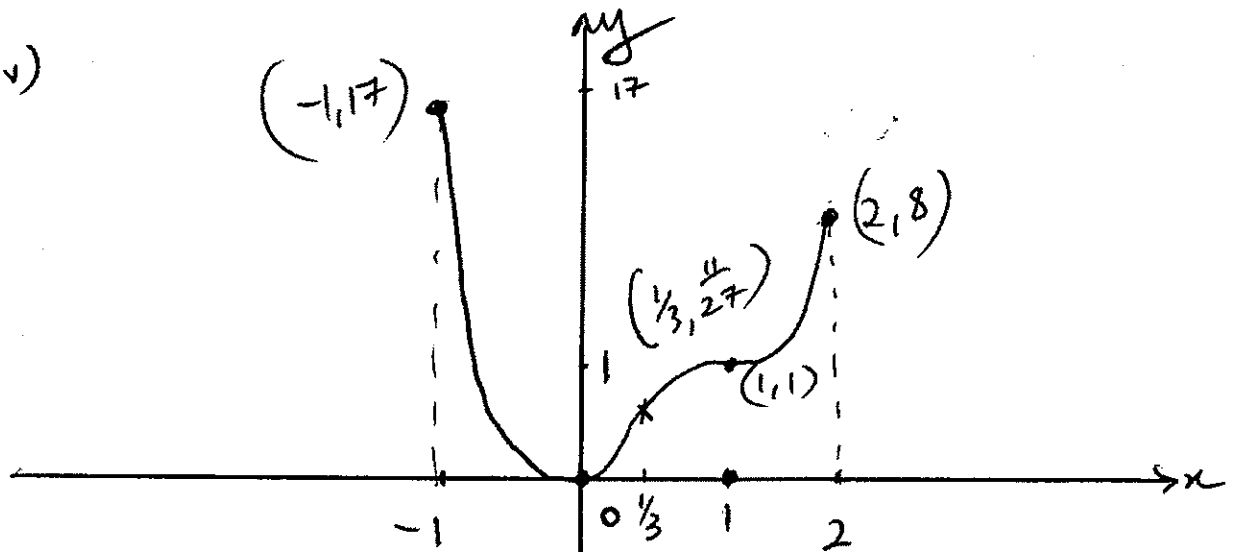
For $x=\frac{1}{3}$

x	$\frac{1}{2}$	1	2
y''	-	0	+

x	$\frac{1}{4}$	$\frac{1}{3}$	$\frac{1}{2}$
y''	+	0	-

(1)

(iv)



end pts ①
showing st. pts ①
general shape showing pt of inflexion ①

QUESTION 3

(a) (i) $\frac{3x^6}{6} - \frac{2x^3}{3} - \frac{x^2}{2} + C$ ①

= $\frac{x^6}{2} - \frac{2}{3}x^3 - \frac{x^2}{2} + C$ ①

(ii) $\frac{3(3x-6)^5}{5 \times 3}$ ①

= $\frac{(3x-6)^5}{5} + C$ ①

$$(iii) \frac{d}{dx} = 2(x)^{-\frac{1}{2}} + 2x^{\frac{1}{2}} - 3 \quad (1)$$

$$= \frac{2x^{\frac{1}{2}}}{\frac{1}{2}} + \frac{2x^{\frac{3}{2}}}{\frac{3}{2}} - 3x + C \quad (1)$$

$$= 4\sqrt{x} + \frac{4x\sqrt{x}}{3} - 3x + C \quad (1)$$

$$(b) \quad \frac{dy}{dx} = \frac{12(x-1)^3}{3} + C$$

$$\frac{dy}{dx} = 4(x-1)^3 + C$$

$$\text{when } \begin{cases} x=1 \\ \frac{dy}{dx}=3 \end{cases}$$

$$3 = 4(1-1)^3 + C \quad (1)$$

$$3 = C$$

$$\frac{dy}{dx} = 4(x-1)^3 + 3 \quad (1)$$

$$y = \frac{\cancel{4}(x-1)^4}{\cancel{4}} + 3x + C \quad \text{when } \begin{cases} x=1 \\ y=2 \end{cases}$$

$$2 = (1-1)^4 + 3 + C$$

$$-1 = C$$

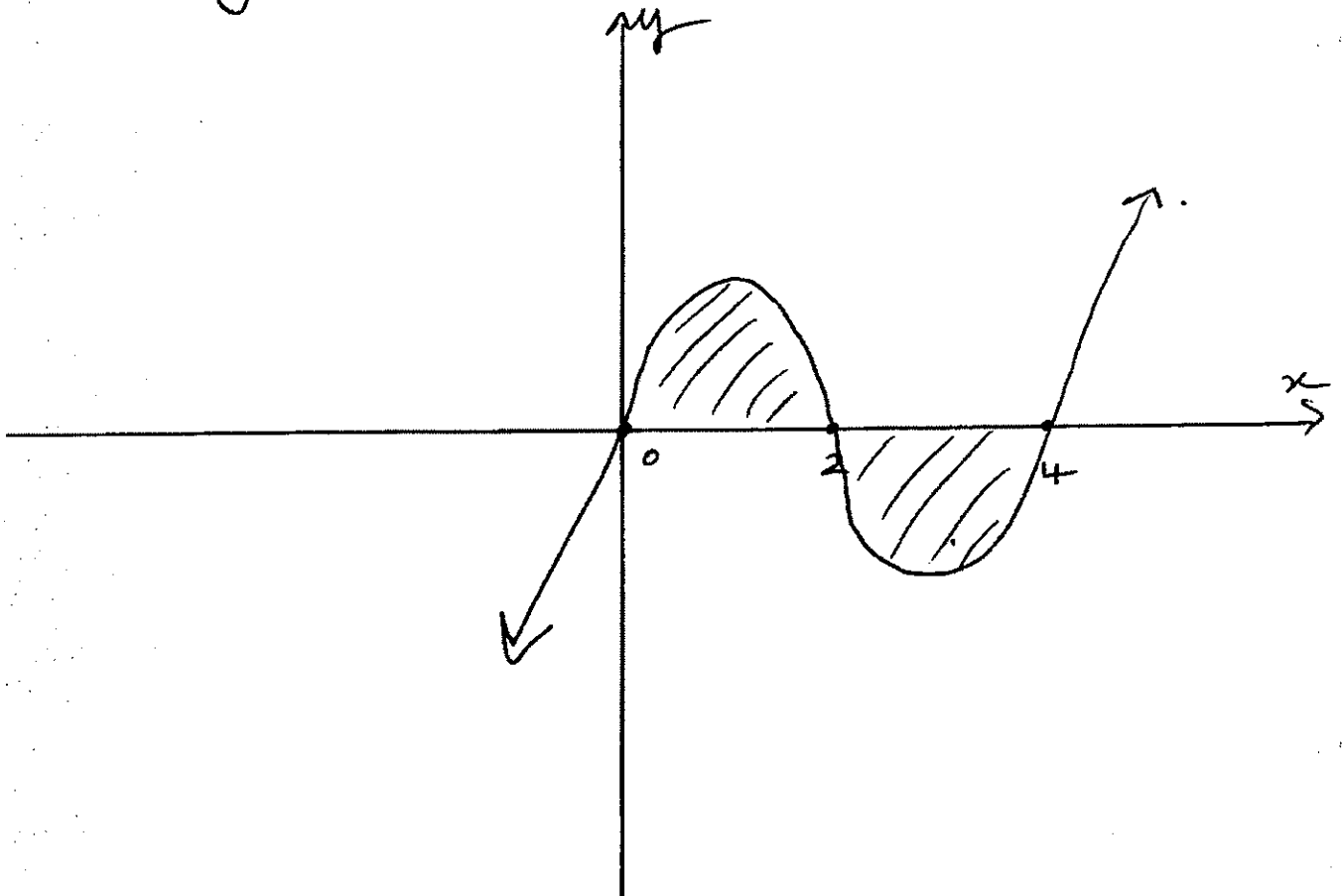
$$\therefore y = (x-1)^4 + 3x - 1 \quad (1)$$

$$(c) (i) A = \int_a^b \underset{\textcircled{1}}{f(x)} dx + \int_b^c \underset{\textcircled{1}}{g(x)} dx$$

$$(ii) A = \int_{-2}^2 \underset{\textcircled{1}}{\sqrt{4-x^2}} dx + \left| \int_2^4 \underset{\textcircled{1}}{-x+2} dx \right|$$

$$(d) \quad y = x(x^2 - 6x + 8)$$

$$y = x(x-2)(x-4)$$



$$\textcircled{1} A = \int_0^2 x^3 - 6x^2 + 8x \, dx + \left| \int_2^4 x^3 - 6x^2 + 8x \, dx \right|$$

$$A = \left[\frac{x^4}{4} - \frac{6x^3}{3} + \frac{8x^2}{2} \right]_0^2 + \left| \left[\frac{x^4}{4} - \frac{6x^3}{3} + \frac{8x^2}{2} \right]_2^4 \right|$$

$$A = \left[\left(\frac{16}{4} - 2 \times 8 + 16 \right) - (0) \right] + \left| \left[\left(\frac{256}{4} - \frac{128}{1} + 64 \right) - \left(\frac{16}{4} - 16 + 16 \right) \right] \right|$$

$$A = 4 + \left| [(0) - (4)] \right|$$

$$A = 4 + 4$$

$$A = 8 \text{ units}^2 \quad \textcircled{1}$$

QUESTION 4:

$$(a) \quad V = \pi \int_{-3}^3 (\sqrt{9-x^2})^2 dx \quad (1)$$

$$V = \pi \int_{-3}^3 9-x^2 dx$$

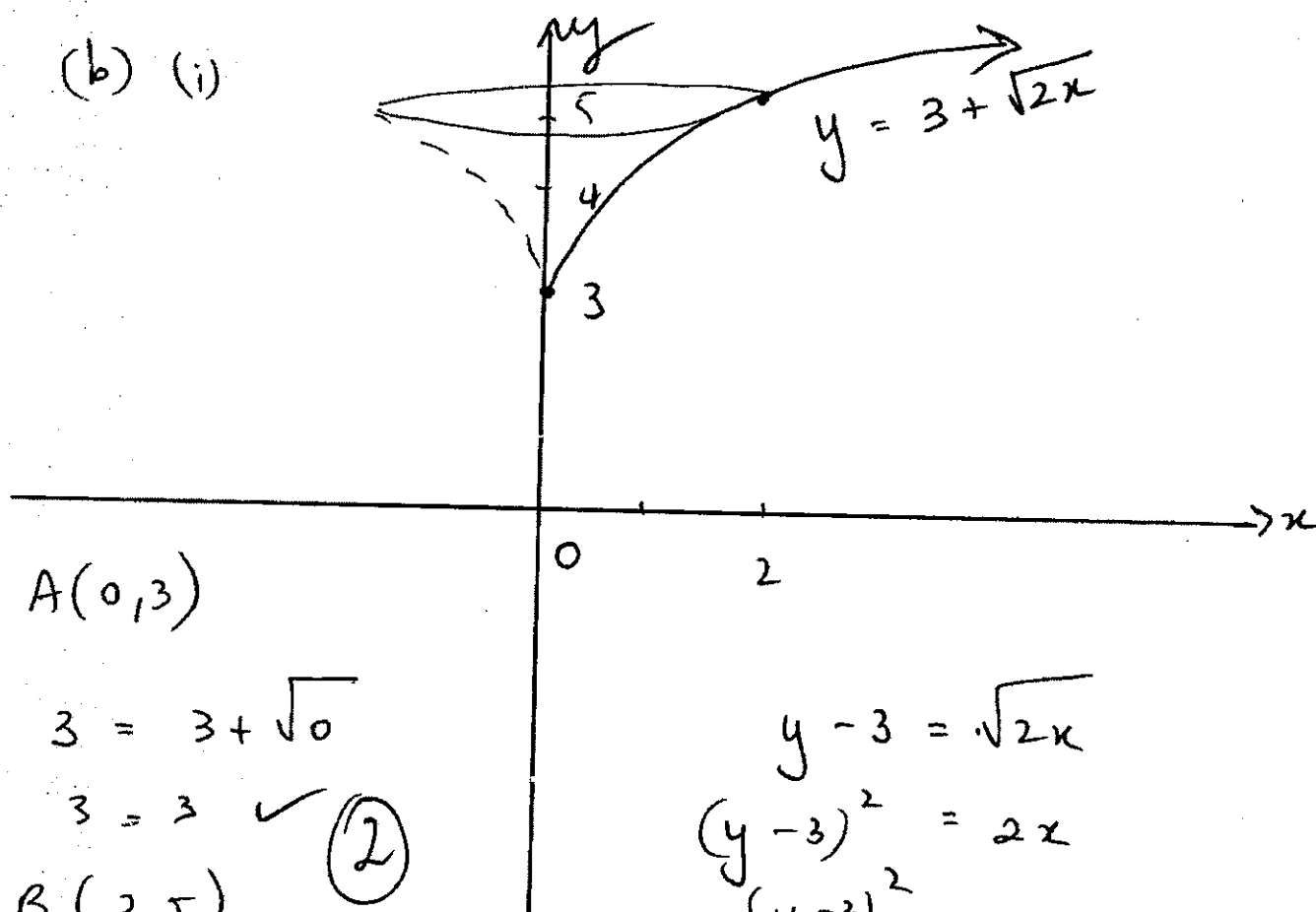
$$V = \pi \left[9x - \frac{x^3}{3} \right]_{-3}^3 \quad (1)$$

$$V = \pi \left[\left(27 - \frac{27}{3} \right) - \left(-27 + \frac{27}{3} \right) \right]$$

$$V = \pi [18 - (-18)]$$

$$V = 36\pi \text{ units}^3 \quad (1)$$

(b) (i)



A(0,3)

$$3 = 3 + \sqrt{0}$$

$$3 = 3 \quad \checkmark \quad (2)$$

B(2,5)

$$5 = 3 + \sqrt{2 \times 2}$$

$$5 = 3 + 2 \quad \checkmark$$

$$\begin{aligned} y - 3 &= \sqrt{2x} \\ (y - 3)^2 &= 2x \\ \frac{(y - 3)^2}{2} &= x \end{aligned}$$

(ii)

$$V = \pi \int_3^5 \frac{(y-3)^4}{4} dy \quad (1)$$

$$V = \pi \left[\frac{(y-3)^5}{20} \right]_3^5 \quad (1)$$

$$V = \pi \left[\frac{2^5}{20} - \frac{0}{20} \right] = \pi \left[\frac{32}{20} \right] = \frac{8}{5} \pi u^3 \quad (1)$$

QUESTION 5:

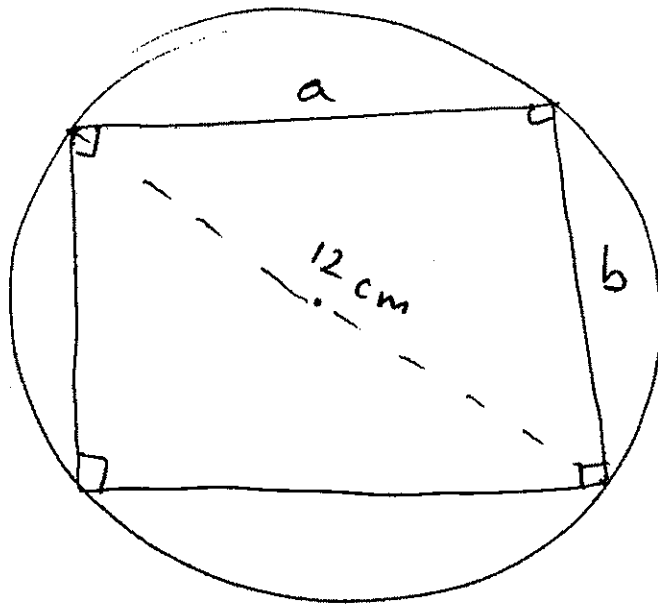
$$(a) \int_0^4 x^{3/2} + x^3 dx \quad (1)$$

$$= \left[\frac{2x^{5/2}}{5} + \frac{x^4}{4} \right]_0^4 \quad (1)$$

$$= \left[\left(\frac{64}{5} + 64 \right) - (0) \right]$$

$$= \frac{384}{5} \text{ OR } 76 \frac{4}{5} \quad (1)$$

(b) (i)



$$a^2 + b^2 = 144$$

$$b^2 = 144 - a^2$$

$$b = \sqrt{144 - a^2}$$

(1)

$$V \text{ of pyramid} = \frac{a \times b \times 18}{3}$$

$$V = \frac{a \times \sqrt{144 - a^2} \times 18}{3}$$

$$V = 6a\sqrt{144 - a^2}$$

$$V = 6\sqrt{a^2(144 - a^2)}$$

(1)

$$V = 6\sqrt{144a^2 - a^4}$$

$$(ii) \quad \frac{dv}{da} = \frac{1}{2} \times 6 (144a^2 - a^4)^{-\frac{1}{2}} \times (288a - 4a^3)$$

$$\text{let } \frac{dv}{da} = 0 \quad 0 = \frac{3(288a - 4a^3)}{(144a^2 - a^4)^{\frac{1}{2}}}$$

$$0 = 3(288a - 4a^3) \quad (1)$$

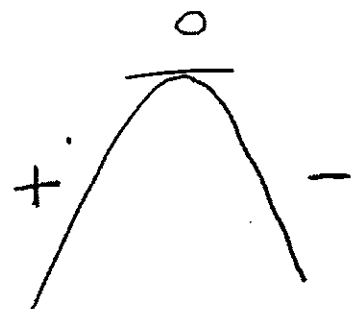
$$0 = 4a(72 - a^2)$$

$$0 = 4a(6\sqrt{2} - a)(6\sqrt{2} + a)$$

$$\therefore \boxed{a = 6\sqrt{2}} \quad (1)$$

a	8	$6\sqrt{2}$	9
$\frac{dv}{da}$	+	0	-

(1)



$$\begin{aligned} V \text{ of pyramid} &= \frac{6\sqrt{2} \times 6\sqrt{2} \times 18}{3} \\ &= 432 \text{ cm}^3 \end{aligned}$$

$$V \text{ of cylinder} = \pi \times 6^2 \times 18$$

$$= 648\pi \text{ cm}^3$$

$\therefore$

$$432 : 648\pi$$

$$2 : 3\pi$$

(1)